

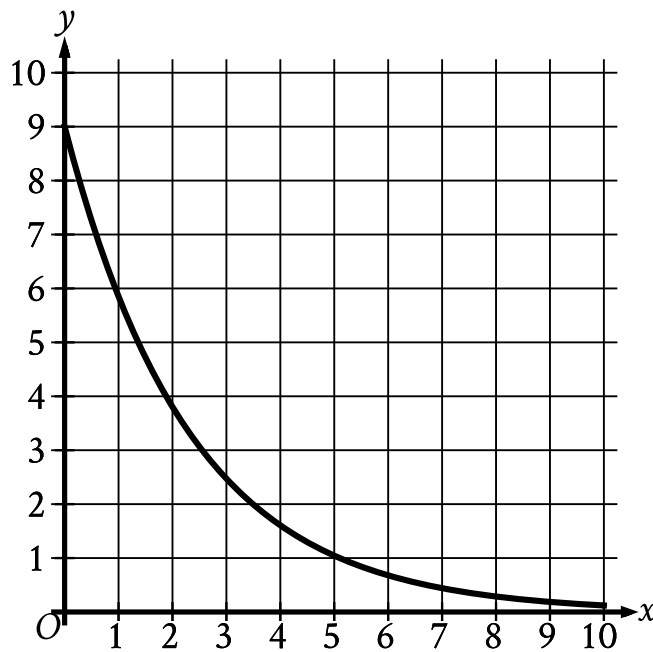
● MEDIUM

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

08



- Moving from left to right:
 - The curve is in quadrant 1.
 - From point (0 comma 9), the curve trends down sharply.
 - From point (5 comma 1), the curve trends down gradually.

The graph gives the estimated number of catalogs y , in thousands, a company sent to its customers at the end of each year, where x represents the number of years since the end of 1992, where $0 \leq x \leq 10$. Which statement is the best interpretation of the y -intercept in this context?

- A. The estimated total number of catalogs the company sent to its customers during the first 10 years was 9,000.
- B. The estimated total number of catalogs the company sent to its customers from the end of 1992 to the end of 2002 was 90.

- C. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9.
- D. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9,000.

Q2

ADVANCED MATH

x	f(x)
0	5
1	$\frac{5}{2}$
2	$\frac{5}{4}$
3	$\frac{5}{8}$

The table above gives the values of the function f for some values of x . Which of the following equations could define f ?

- A. $f(x) = 5(2^{x+1})$
- B. $f(x) = 5(2^x)$
- C. $f(x) = 5(2^{-(x+1)})$
- D. $f(x) = 5(2^{-x})$

Q3

GRID-IN ADVANCED MATH

The function f is defined by $fx = 8x^3 + 4$. What is the value of $f2$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$f(t) = 5000.5^{\frac{t}{12}}$$

The function f models the intensity of an X-ray beam, in number of particles in the X-ray beam, t millimeters below the surface of a sample of iron. According to the model, what is the estimated number of particles in the X-ray beam when it is at the surface of the sample of iron?

- A. 500
- B. 12
- C. 5
- D. 2

Q5

GRID-IN

ADVANCED MATH

$$f(x) = x^5 + 9x + 17$$

For the given function f , the graph of $y = f(x)$ in the xy -plane passes through the point $(0, b)$, where b is a constant. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Time (years)	Total amount (dollars)
0	670.00
1	674.02
2	678.06

Sara opened a savings account at a bank. The table shows the exponential relationship between the time t , in years, since Sara opened the account and the total amount d , in dollars, in the account. If Sara made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and d ?

- A. $d = 0.0061 + 670^t$
- B. $d = 6701 + 0.006^t$
- C. $d = 1 + 0.006^t$
- D. $d = 1 + 670^t$

Q7

ADVANCED MATH

According to Moore's law, the number of transistors included on microprocessors doubles every 2 years. In 1985, a microprocessor was introduced that had 275,000 transistors. Based on this information, in which of the following years does Moore's law estimate the number of transistors to reach 1.1 million?

- A. 1987
- B. 1989
- C. 1991
- D. 1994

Q8

ADVANCED MATH

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 3,364 feet above the ground when it is dropped at $t = 0$. Approximately how many seconds after being dropped does the function estimate the object will hit the ground?

- A. 7.25
- B. 14.50
- C. 105.13
- D. 210.25

$$S(n) = 38,000a^n$$

The function S above models the annual salary, in dollars, of an employee n years after starting a job, where a is a constant. If the employee's salary increases by 4% each year, what is the value of a ?

- A. 0.04
- B. 0.4
- C. 1.04
- D. 1.4

The exponential function g is defined by $g(x) = 19 \cdot a^x$, where a is a positive constant. If $g(3) = 2,375$, what is the value of $g(4)$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.